BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
Department of Electrical and Electronic Engineering

PROJECT REPORT

CMOS Inverter Characterization

Study of the Voltage Transfer Characteristic using Discrete MOSFETs

Course No.	EEE 202 (H)
Course Title	Electronic Circuits I Laboratory (Hardware)
Project Type	Hardware Implementation and PSpice Simulation

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Date: May 2026

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Abstract

This project investigates the operation and voltage transfer characteristic of a CMOS inverter implemented using discrete MOSFETs. The hardware circuit used an IRF9540 PMOS transistor and an IRF3710 NMOS transistor with a 5 V DC supply. The input voltage was varied from 0 V to 5 V, and the corresponding output voltage was measured at the common drain node. The experimental VTC showed a sharp HIGH-to-LOW transition near V_{in} = 3.0 V, and the switching voltage was determined as V_M = 2.996 V. A PSpice DC sweep simulation using generic MbreakP and MbreakN models was also performed to verify the qualitative inverter behavior. The simulation produced the expected inverting characteristic with a switching point close to the mid-supply voltage.

1. Objectives

1. To study the operation of a CMOS inverter using one PMOS and one NMOS transistor.
2. To verify experimentally that the output voltage is the logical complement of the input voltage.
3. To obtain the voltage transfer characteristic (VTC) of the inverter.
4. To determine the switching voltage from the plotted VTC.
5. To compare the experimental behavior with a PSpice DC sweep simulation.

2. Components and Equipment

Component / Equipment	Specification / Use
PMOS transistor	IRF9540
NMOS transistor	IRF3710
DC supply	V_{DD} = 5 V
Variable input source	V_{in} swept from 0 V to 5 V
Measurement instrument	Digital multimeter
Breadboard and wires	Circuit implementation
LED and resistor	Optional visual output indicator; 1 kΩ series resistor
Software	PSpice (DC sweep simulation); MATLAB (plotting)

3. Theory

A CMOS inverter is a fundamental digital logic circuit formed by connecting one PMOS transistor and one NMOS transistor in a complementary configuration. The source terminal of the PMOS is connected to the supply voltage V_{DD}, while the source terminal of the NMOS is connected to ground. The gates of both transistors are tied together to form the input node V_{in}, and the drains are tied together to form the output node V_{out}.

When V_{in} is low, the PMOS transistor turns ON and the NMOS transistor turns OFF. As a result, the output node is pulled up toward V_{DD}, producing a HIGH output. When V_{in} is high, the PMOS turns OFF and the NMOS turns ON, so the output node is pulled down to ground, producing a LOW output. Therefore, the output is the logical complement of the input.

Near the switching region, both transistors may conduct partially. Neglecting channel-length modulation, the approximate drain-current expressions in saturation are:

$$I_{DN} = \tfrac{1}{2}\,\beta_n\,(V_{in} - V_{TN})^2$$

$$I_{DP} = \tfrac{1}{2}\,\beta_p\,(V_{DD} - V_{in} - |V_{TP}|)^2$$

At the switching point V_M, the two currents are equal, $I_{DN} = I_{DP}$.

For a nearly symmetrical inverter, the switching voltage is expected to be close to $V_{DD}/2$. In practical discrete-transistor circuits, the switching point can shift due to threshold-voltage mismatch and device non-idealities.

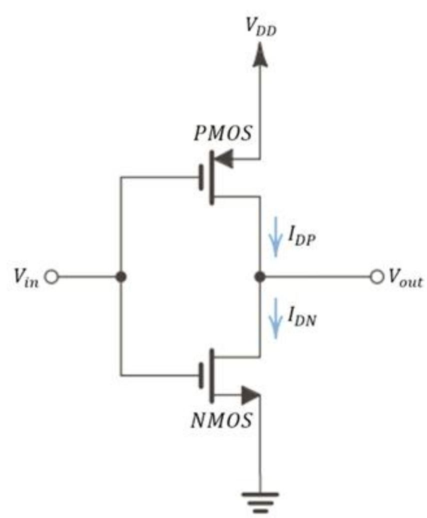

Figure 1. *Circuit diagram of the CMOS inverter.*

4. Circuit Configuration and Hardware Setup

Node / Terminal	Hardware Connection
IRF9540 PMOS source	V_{DD} = 5 V
IRF9540 PMOS gate	V_{in}
IRF9540 PMOS drain	V_{out}
IRF3710 NMOS drain	V_{out}
IRF3710 NMOS gate	V_{in}
IRF3710 NMOS source	GND

The experimental setup used an LED with a 1 kΩ resistor as a visual indicator during logic verification. However, the measured VTC data reported in this document was taken under the unloaded output condition, so the PSpice simulation was also performed without an output load.

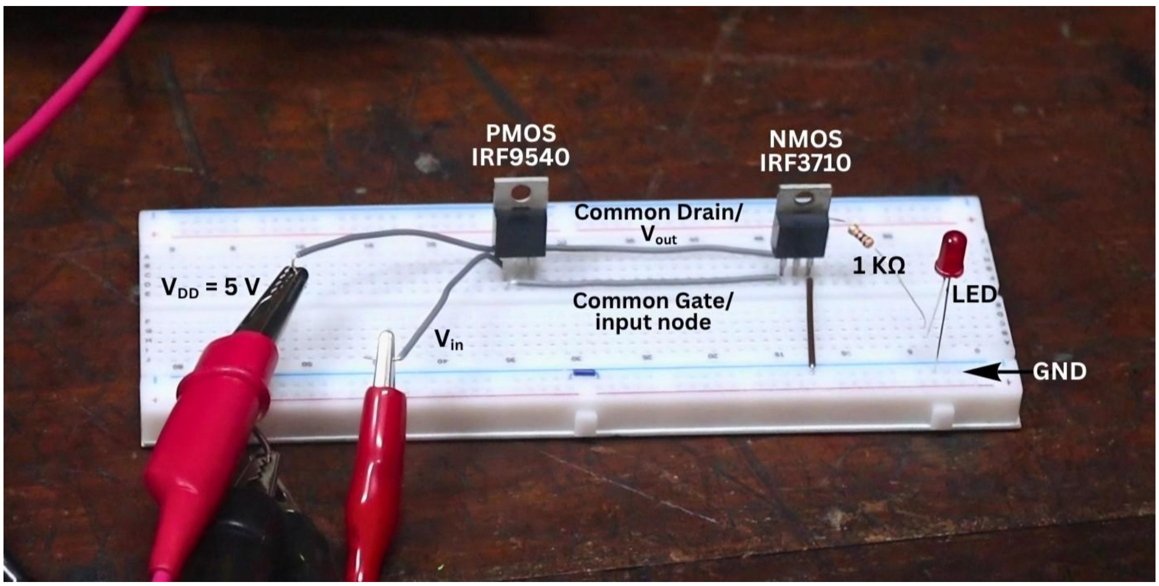

Figure 2. *Practical breadboard implementation of the CMOS inverter.*

5. Experimental Procedure

1. The CMOS inverter circuit was assembled on the breadboard according to the complementary PMOS–NMOS configuration.
2. A fixed supply voltage V_{DD} = 5 V was applied to the PMOS source, while the NMOS source was connected to ground.
3. The input voltage V_{in} was applied to the common gate node of the PMOS and NMOS transistors.
4. The output voltage V_{out} was measured at the common drain node.
5. V_{in} was varied gradually from 0 V to 5 V, and the corresponding V_{out} was recorded.
6. Finer input increments were used near the transition region to capture the sharp switching behavior.
7. The measured data was plotted as V_{out} versus V_{in}, and the switching voltage was determined from the intersection with the line $V_{out} = V_{in}$.

6. Experimental Data and Results

Sl.	V_{in} (V)	V_{out} (V)	Sl.	V_{in} (V)	V_{out} (V)
1	0.0	4.82	11	3.0	2.90
2	0.5	4.82	12	3.1	0.00
3	1.0	4.82	13	3.2	0.00
4	1.5	4.82	14	3.3	0.00
5	2.0	4.82	15	3.4	0.00
6	2.5	4.82	16	3.5	0.00
7	2.6	4.82	17	4.0	0.00
8	2.7	4.82	18	4.5	0.00
9	2.8	4.81	19	5.0	0.00
10	2.9	4.70			

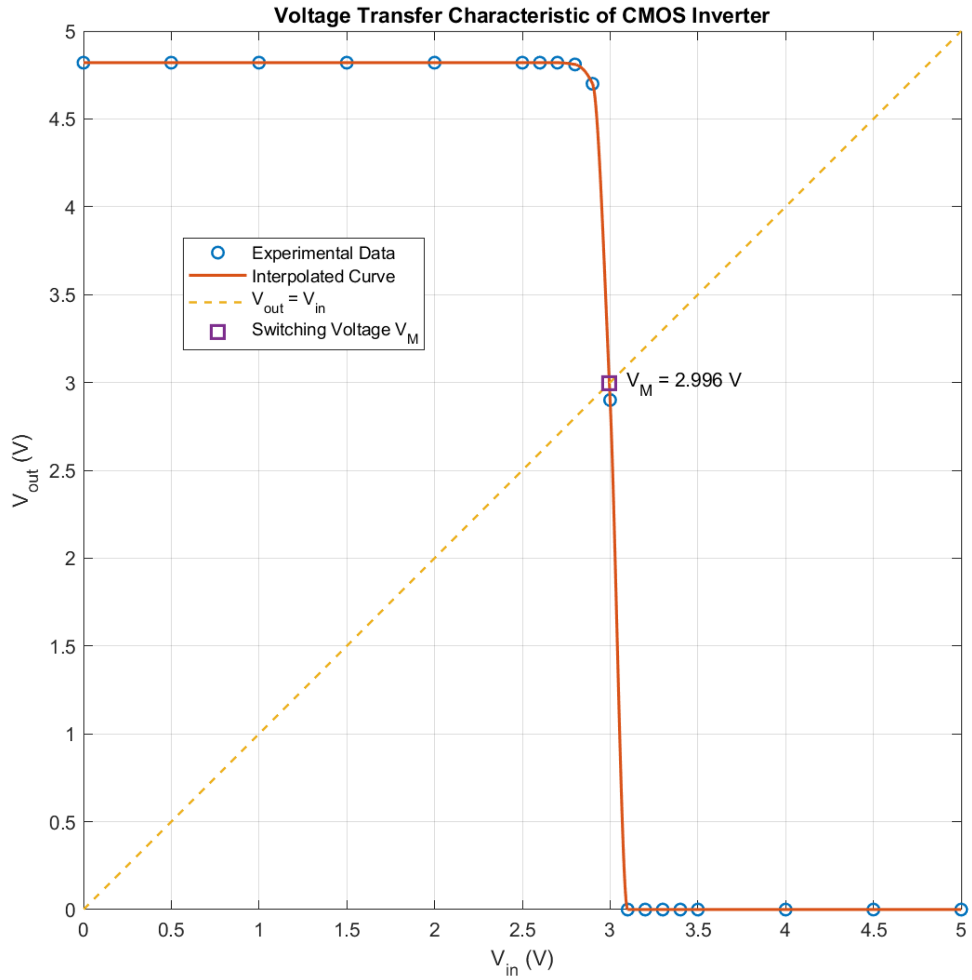

Figure 3. *Experimental voltage transfer characteristic of the CMOS inverter.*

From the plotted experimental VTC, the switching voltage was obtained from the point where the inverter characteristic intersects the line $V_{out} = V_{in}$. The measured switching voltage was:

$$V_M = 2.996 \text{ V} \approx 3.0 \text{ V}$$

7. PSpice Simulation

A PSpice DC sweep simulation was performed to verify the qualitative switching behavior of the CMOS inverter. The simulation used an unloaded inverter configuration to match the unloaded experimental VTC measurement. Generic MbreakP and MbreakN MOSFET models were used. Therefore, the simulated switching voltage is intended for qualitative comparison, not exact device-level matching with the IRF9540 and IRF3710 hardware circuit.

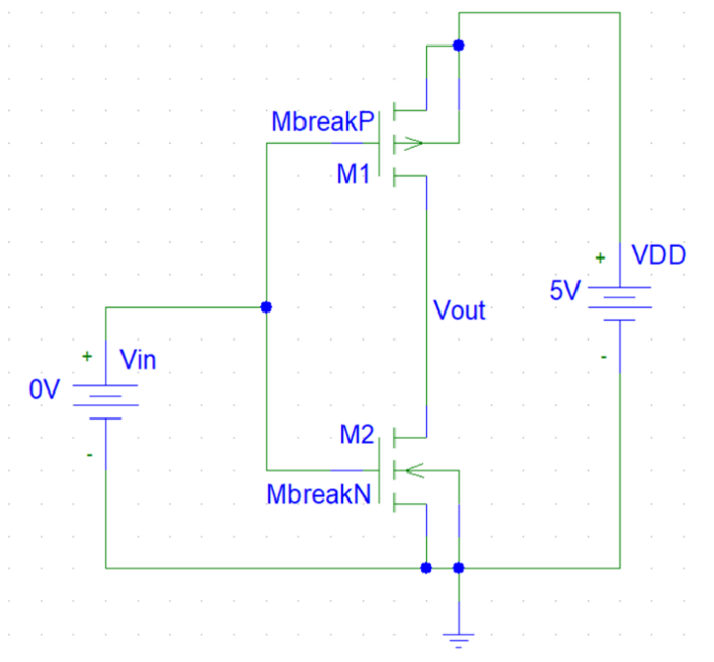

Figure 4. *PSpice schematic used for the DC sweep simulation.*

Simulation Parameter	Value
Analysis type	DC Sweep
Sweep source	V_{in}
Sweep range	0 V to 5 V
Step size	0.01 V
Supply voltage	V_{DD} = 5 V
Output node	V_{out} at common drain
Output load	No load

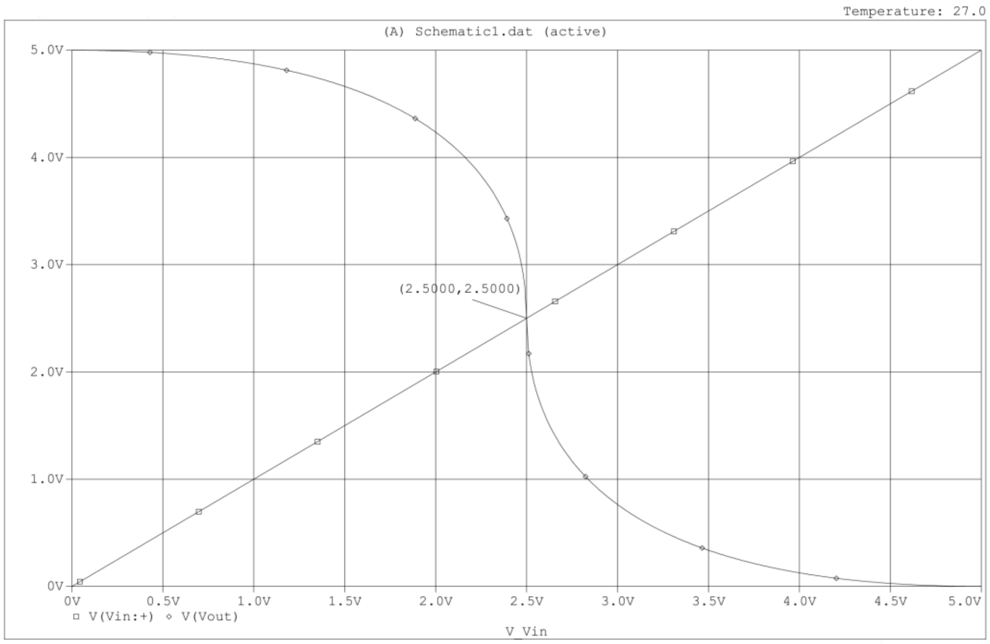

Figure 5. *Simulated voltage transfer characteristic from the PSpice DC sweep analysis.*

8. Experimental and Simulation Comparison

Quantity	Experimental Result	PSpice Simulation
Supply voltage	5 V	5 V
MOSFET model	IRF9540 PMOS, IRF3710 NMOS	Generic MbreakP, MbreakN
Output condition	Unloaded during VTC measurement	Unloaded
HIGH output level	≈ 4.82 V	≈ 5 V
LOW output level	0.00 V	≈ 0 V
Switching voltage	V_M = 2.996 V ≈ 3.0 V	≈ 2.5 V

The experimental and simulated characteristics both confirm the inverting behavior of the CMOS inverter. The simulated curve switches close to $V_{DD}/2$ because the generic Mbreak models are comparatively symmetrical. In contrast, the practical circuit switches near 3.0 V because the discrete PMOS and NMOS devices are not perfectly matched. Threshold-voltage mismatch, power-MOSFET characteristics, breadboard contact resistance, parasitic effects, and measurement limitations can all shift the practical switching point.

9. Explanation of Operation

9.1 Low Input Voltage

For low V_{in}, the PMOS transistor remains ON and the NMOS transistor remains OFF. Therefore, V_{out} is pulled up toward V_{DD}. In the experiment, the output remained around 4.82 V for low input values, which represents logic HIGH.

9.2 High Input Voltage

For high V_{in}, the PMOS transistor turns OFF and the NMOS transistor turns ON. As a result, V_{out} is pulled down to ground. In the experiment, for $V_{in} \geq 3.1$ V, the output became 0 V, representing logic LOW.

9.3 Intermediate Input Voltage

For intermediate V_{in}, both transistors conduct partially and the output changes rapidly from HIGH to LOW. In this experiment, the transition occurred sharply around $V_{in} = 3.0$ V.

Input Condition	Measured V_{in}	Measured V_{out}	Logic Output
LOW	0 V	4.82 V	HIGH (1)
HIGH	5 V	0.00 V	LOW (0)

10. Discussion

The experimental results confirmed the correct operation of the CMOS inverter. For low input voltages, the output remained close to the supply voltage, while for high input voltages, the output dropped to zero. The VTC showed a sharp transition region between approximately 2.9 V and 3.1 V.

The switching voltage was found to be 2.996 V, which is higher than the ideal mid-supply value of 2.5 V. This difference is expected because the implemented circuit used discrete power MOSFETs rather than matched CMOS transistors on an integrated circuit. The actual threshold voltages and transfer characteristics of the IRF9540 and IRF3710 devices are not symmetrical.

The PSpice simulation using generic Mbreak models produced a switching point close to 2.5 V, which supports the theoretical behavior of a symmetrical CMOS inverter. The difference between the simulated and experimental switching points illustrates the effect of practical device mismatch and real hardware limitations.

11. Conclusion

A CMOS inverter was successfully implemented using an IRF9540 PMOS transistor and an IRF3710 NMOS transistor. The experimental results verified the complementary switching action of the PMOS and NMOS devices. The measured voltage transfer characteristic showed a sharp transition from HIGH to LOW, and the switching voltage was determined as $V_M = 2.996$ V, approximately 3.0 V.

PSpice DC sweep simulation using generic MOSFET models further confirmed the expected inverting behavior. The project demonstrates both the practical operation and the voltage transfer characteristic of a CMOS inverter, while also showing how real device mismatch can shift the switching voltage from the ideal mid-supply value.

References

1. A. S. Sedra and K. C. Smith, Microelectronic Circuits, Oxford University Press.
2. IRF9540 P-Channel Power MOSFET datasheet.
3. IRF3710 N-Channel Power MOSFET datasheet.
4. PSpice documentation and generic breakout MOSFET models.

Appendix: Repository Structure

Folder / File	Purpose
report/	PDF report
hardware/	Breadboard setup image and circuit reference images
simulation/	PSpice schematic screenshot and simulated VTC plot
data/	Measured VTC data in CSV format
analysis/	Plotting script used to generate the experimental VTC curve
docs/	Theory, methodology, and reference notes